

Assignment 2 : Product Inventory Management (CRUD Only)

Database Setup

Database: storeDB

Collection: products

```
> db.createCollection('product')  
< { ok: 1 }
```

Task 1: Insert Documents :

Insert at least 7 product records with these fields:

```
17  
< {  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('68e7f88f6e9e24c343d0a672'),  
    '1': ObjectId('68e7f88f6e9e24c343d0a673'),  
    '2': ObjectId('68e7f88f6e9e24c343d0a674'),  
    '3': ObjectId('68e7f88f6e9e24c343d0a675'),  
    '4': ObjectId('68e7f88f6e9e24c343d0a676'),  
    '5': ObjectId('68e7f88f6e9e24c343d0a677'),  
    '6': ObjectId('68e7f88f6e9e24c343d0a678')  
  }  
}
```

Task 2: Basic Queries (FIND)

Find all products

```
> db.product.find()
< {
  _id: ObjectId('68e7f88f6e9e24c343d0a672'),
  pid: 1,
  name: 'iPhone 15 Pro',
  category: 'Electronics',
  price: 1299.99,
  stock: 50,
  brand: 'Apple',
  addedDate: '2025-10-01'
}
{
  _id: ObjectId('68e7f88f6e9e24c343d0a673'),
  pid: 2,
  name: 'Nike Air Max',
  category: 'Footwear',
  price: 199.99,
  stock: 120,
  brand: 'Nike',
  addedDate: '2025-09-15'
}
{
  _id: ObjectId('68e7f88f6e9e24c343d0a674'),
  pid: 3,
  name: 'Samsung Galaxy S24',
  category: 'Electronics',
  price: 1099,
  stock: 70,
  brand: 'Samsung',
  addedDate: '2025-08-20T00:00:00.000Z'
}
```

Find products in category "Electronics"

```
> db.product.find({category: 'Electronics'})
< {
  _id: ObjectId('68e7f88f6e9e24c343d0a672'),
  pid: 1,
  name: 'iPhone 15 Pro',
  category: 'Electronics',
  price: 1299.99,
  stock: 50,
  brand: 'Apple',
  addedDate: '2025-10-01'
}
{
  _id: ObjectId('68e7f88f6e9e24c343d0a674'),
  pid: 3,
  name: 'Samsung Galaxy S24',
  category: 'Electronics',
  price: 1099,
  stock: 70,
  brand: 'Samsung',
  addedDate: '2025-08-20T00:00:00.000Z'
}
```

Find products with price greater than 2000

```
> db.product.find({price:{$gt:1000}})
< {
  _id: ObjectId('68e7f88f6e9e24c343d0a672'),
  pid: 1,
  name: 'iPhone 15 Pro',
  category: 'Electronics',
  price: 1299.99,
  stock: 50,
  brand: 'Apple',
  addedDate: '2025-10-01'
}
{
  _id: ObjectId('68e7f88f6e9e24c343d0a674'),
  pid: 3,
  name: 'Samsung Galaxy S24',
  category: 'Electronics',
  price: 1099,
  stock: 70,
  brand: 'Samsung',
  addedDate: 2025-08-20T00:00:00.000Z
}
```

Find one product from "Furniture" category

```
> db.product.findOne({category:'Footwear'})
< {
  _id: ObjectId('68e7f88f6e9e24c343d0a673'),
  pid: 2,
  name: 'Nike Air Max',
  category: 'Footwear',
  price: 199.99,
  stock: 120,
  brand: 'Nike',
  addedDate: '2025-09-15'
}
```

Task 3: Updates

Update price of "Nike Air Max" to 10000

```
> db.product.updateOne({name:'Nike Air Max'},{$set:{price:10000}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.product.find({name:'Nike Air Max'})
< {
  _id: ObjectId('68e7f88f6e9e24c343d0a673'),
  pid: 2,
  name: 'Nike Air Max',
  category: 'Footwear',
  price: 10000,
  stock: 120,
  brand: 'Nike',
  addedDate: '2025-09-15'
}
```

Change category of "Computers" to "Electronics"

```
> db.product.updateOne({category:'Computers'},{$set:{category:'Electronics'}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
> db.product.find({category: 'Electronics'})
```

```
< {  
  _id: ObjectId('68e7f88f6e9e24c343d0a678'),  
  pid: 7,  
  name: 'Dell XPS 15',  
  category: 'Electronics',  
  price: 1899,  
  stock: 35,  
  brand: 'Dell',  
  addedDate: 2025-06-18T00:00:00.000Z  
}
```

Update stock of "Sony" to 70

```
> db.product.updateOne({brand: 'Sony'}, {$set: {stock: 70}})
```

```
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

```
> db.product.find({brand: 'Sony'})
```

```
< {  
  _id: ObjectId('68e7f88f6e9e24c343d0a676'),  
  pid: 5,  
  name: 'Sony WH-1000XM5',  
  category: 'Electronics',  
  price: 399.99,  
  stock: 70,  
  brand: 'Sony',  
  addedDate: 2025-07-12T00:00:00.000Z  
}
```

Task 4: Deletions

Delete one product with price less than 1000

```
> db.product.deleteOne({price:{$lte:1000}})
< {
  acknowledged: true,
  deletedCount: 1
}
> db.product.find({price:1000})
<
> db.product.find({price:399.99})
< {
  _id: ObjectId('68e7f88f6e9e24c343d0a676'),
  pid: 5,
  name: 'Sony WH-1000XM5',
  category: 'Electronics',
  price: 399.99,
  stock: 70,
  brand: 'Sony',
  addedDate: 2025-07-12T00:00:00.000Z
}
```

Delete all products from category "Electronics"

```
> db.product.deleteMany({category:'Electronics'})
< {
  acknowledged: true,
  deletedCount: 4
}
> db.product.find()
< {
  _id: ObjectId('68e7f88f6e9e24c343d0a673'),
  pid: 2,
  name: 'Nike Air Max',
  category: 'Footwear',
  price: 10000,
  stock: 120,
  brand: 'Nike',
  addedDate: '2025-09-15'
}
{
  _id: ObjectId('68e7f88f6e9e24c343d0a677'),
  pid: 6,
  name: 'Adidas Ultraboost',
  category: 'Footwear',
  price: 180,
  stock: 140,
  brand: 'Adidas',
}
```

2.Hospital Patients Records (CRUD Only)

Task 1: Insert Documents Insert at least 7 patient records with fields: pid (Patient ID), name, age, disease, doctor, city, admissionDate:

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('68e75a2056f0f4ff8ae738ab'),
    '1': ObjectId('68e75a2056f0f4ff8ae738ac'),
    '2': ObjectId('68e75a2056f0f4ff8ae738ad'),
    '3': ObjectId('68e75a2056f0f4ff8ae738ae'),
    '4': ObjectId('68e75a2056f0f4ff8ae738af'),
    '5': ObjectId('68e75a2056f0f4ff8ae738b0'),
    '6': ObjectId('68e75a2056f0f4ff8ae738b1')
  }
}
```

Task 2: Basic Queries (Find):

Find all patients:

```
> db.patients.find()
< {
  _id: ObjectId('68e75a2056f0f4ff8ae738ab'),
  pid: '101',
  name: 'Rahul Sharma',
  age: 32,
  disease: 'Fever',
  doctor: 'Dr. Mehta',
  city: 'Mumbai',
  admissionDate: 2025-10-01T00:00:00.000Z
}
{
  _id: ObjectId('68e75a2056f0f4ff8ae738ac'),
  pid: '102',
  name: 'Sneha Patel',
  age: 27,
  disease: 'Asthma',
  doctor: 'Dr. Reddy',
  city: 'Ahmedabad',
  admissionDate: 2025-09-25T00:00:00.000Z
}
```

Find patients treated by "Dr. Iyer":

```
> db.patients.find({ doctor: "Dr. Iyer" })
< {
  _id: ObjectId('68e75a2056f0f4ff8ae738b1'),
  pid: '107',
  name: 'Karan Joshi',
  age: 50,
  disease: 'Heart Disease',
  doctor: 'Dr. Iyer',
  city: 'Pune',
  admissionDate: 2025-09-28T00:00:00.000Z
}
```


Find patients older than 40:

```
> db.patients.find({ age: { $gt: 40 } })
< {
  _id: ObjectId('68e75a2056f0f4ff8ae738ad'),
  pid: '103',
  name: 'Amit Verma',
  age: 45,
  disease: 'Diabetes',
  doctor: 'Dr. Kapoor',
  city: 'Delhi',
  admissionDate: 2025-09-18T00:00:00.000Z
}
{
  _id: ObjectId('68e75a2056f0f4ff8ae738b1'),
  pid: '107',
  name: 'Karan Joshi',
  age: 50,
  disease: 'Heart Disease',
  doctor: 'Dr. Iyer',
  city: 'Pune',
  admissionDate: 2025-09-28T00:00:00.000Z
}
```

Find one patient with disease "Diabetes":

```
> db.patients.findOne({ disease: "Diabetes" })
< {
  _id: ObjectId('68e75a2056f0f4ff8ae738ad'),
  pid: '103',
  name: 'Amit Verma',
  age: 45,
  disease: 'Diabetes',
  doctor: 'Dr. Kapoor',
  city: 'Delhi',
  admissionDate: 2025-09-18T00:00:00.000Z
}
```

Task 3: Updates

Update disease of "Amit Verma" to "Fever":

```
> db.patients.updateOne(
  { name: "Amit Verma" },
  { $set: { disease: "Fever" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Change doctor of "Karan Joshi" to "Dr. Singh":

```
> db.patients.updateOne(
  { name: "Karan Joshi" },
  { $set: { doctor: "Dr. Singh" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Update city of "Amit Verma" to "Delhi" :

```
> db.patients.updateOne(
  { name: "Amit Verma" },
  { $set: { city: "Delhi" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
```

Task 4: Deletions

Delete one patient with age less than 25:

```
> db.patients.deleteOne({ age: { $lt: 25 } })  
< {  
  acknowledged: true,  
  deletedCount: 1  
}
```

Delete all patients whose disease is "Cold":

```
> db.patients.deleteMany({ disease: "Cold" })  
< {  
  acknowledged: true,  
  deletedCount: 0  
}
```